Quantum Computing: Start your Journey with Qiskit!

Quantum computing has been the buzzword in 2020 (apart from COVID-19). The global awakening is noticeable, with an active community of thousands working rigorously on honing their skills in this domain. Just the necessary bits of linear algebra and Python are enough to get you started on this path. This article will introduce you to the ways of accessing the quantum advantage, using Qiskit.

Qiskit is an open source library that facilitates the use of quantum properties with Python. It has been designed and developed by IBM Quantum and the general quantum community for people like us who like to experiment.

For this article, we expect our readers to have a basic familiarity with Python, linear algebra and complex numbers along with the theoretical fundamentals of quantum computing. Throughout this tutorial, library imports are done only once the very first requirement. The code will work fine if sequentially reproduced. In case a part of it is being used, the required libraries must be imported along with the code.

Before jumping into the coding part, let us answer a very basic but genuine question that is the most important ingredient of any experiment in the quantum realm. The question is: Why quantum over classical computing? Is it for the fancy circuits? The answer is 'no'. Then, is it the craze of trying out something new? Once again, the answer is 'no'. It must be the weird physics and mathematics driving the motivation perhaps? Well, that could be partly true. Quantum computing gets its bold features from quantum physics, which takes it to an advanced level of performing computations. It can do what classical computers can't. Quantum computing can perform classically expensive computations in an efficient manner. In fact, 'quantum supremacy' is no more just a term given by John Preskill, and was realised by Google's quantum research team in 2019. They proved that a problem that would take approximately 10 thousand years to be solved by the most powerful supercomputer today, was solved by a 53-qubit quantum processor (Sycamore) in feasible time. It proved that quantum computers have the capability of performing in a way that classical computers cannot think of. The weird properties of quantum mechanics have enabled quantum computing to take that 'quantum advantage' over its classical counterpart and be quick about it. With this article, we will introduce you to the ways of accessing the quantum advantage.